Label-Efficient Dark-Vessel Detection in SAR: A Controlled Comparison of Encoder Pretraining

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2026

Abstract

Human-verified labels for vessels without matched Automatic Identification System reports are scarce. We test whether generic ImageNet, optical remote-sensing, or SAR pretraining reduces the labeled-data requirement for point detection in Sentinel-1 imagery. Two size-matched tracks use ViT-B/16 and ConvNeXt-V2-Base backbones. Four initialization roles within each track share the input, detector, schedule, scene subsets, seed, and strict-FP32 execution. The design contains 32 cells across four nested label budgets. The submission snapshot records campaign progress but lacks the completion receipts needed to admit numerical results. We therefore report the design, current status, and fail-closed result path without an H100 ranking or label-efficiency claim. The snapshot contains no test metric; the human-verified set remains sealed.

1 Introduction

Sentinel-1 synthetic-aperture radar (SAR) supplies maritime imagery through clouds and without daylight [7]. Analysts can use it to find vessels that lack a contemporaneous matched Automatic Identification System report. These dark-vessel candidates matter for fisheries enforcement, maritime safety, and sanctions monitoring. They also present a difficult learning problem: a vessel occupies few pixels in a large scene, coastal infrastructure creates strong clutter, and human verification costs time. The xView3-SAR benchmark provides georeferenced point labels and a geographic matching protocol for this setting [3].

Transfer learning may reduce the number of target labels required for a useful detector. Generic ImageNet features offer broad visual structure. Optical remote-sensing checkpoints add overhead-scene statistics, while SAR checkpoints share the sensing modality. Public checkpoints confound source domain with model family and pretraining objective, so an uncontrolled comparison cannot isolate transfer effects.

We use two architecture-matched tracks. Four ViT-B/16 arms and four ConvNeXt-V2-Base arms differ only in initialization within their track: random, ImageNet, optical remote sensing, or SAR. Matched roles across tracks show whether a within-track pattern survives a change in architecture. We make no new detector or pretraining claim. The contribution is a controlled 32-cell design, one corrected evaluation contract, and a provenance-bound H100 result path.

2 Dataset and Evaluation Protocol

The frozen xView3 cohort contains 111 training scenes, 23 development scenes, and 16 test scenes. A separate set of 50 human-verified scenes remains sealed for one final evaluation. Scene identifiers, rather than chips, define every split. The four training budgets contain 12, 28, 56, and 111 scenes and satisfy $S_{10} \subset S_{25} \subset S_{50} \subset S_{100}$. All arms see the same scene set at a given budget.

The preprocessing path converts each dual-polarization GRD scene into a fixed three-channel tensor $[\mathrm{VH}, \mathrm{VV}, \mathrm{VH} - \mathrm{VV}]$ in decibels. It computes normalization statistics from training scenes only and creates overlapping 800-pixel chips. Optimization samples shared 512-pixel crops from these chips.

The corrected label contract treats a HIGH- or MEDIUM-confidence row as a positive only when `is_vessel` is true. HIGH/MEDIUM non-vessels form background; LOW-confidence rows define ignore regions. The

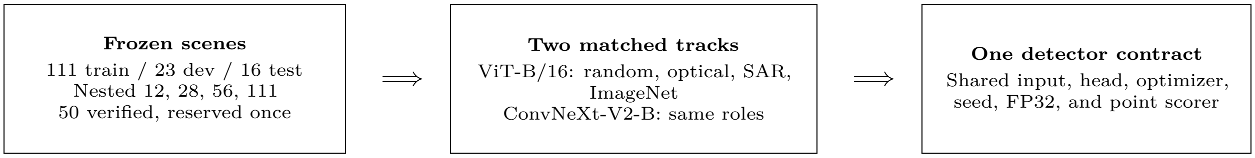

Figure 1: Controlled study design. Initialization is the only variable within each architecture track. Matched roles are compared descriptively across tracks; the final human-verified set remains sealed until the cohort and analysis code are frozen.

Table 1: Controlled initialization roles. Source heads are discarded.

Role	ViT-B/16	ConvNeXt-V2-Base
Random	fresh initialization	fresh initialization
Optical	SatDINO, fMoW-RGB	BigEarthNet Sentinel-2
SAR	SARMAE, SAR-1M	BigEarthNet Sentinel-1
ImageNet	supervised AugReg	FCMAE + supervised

eight development scenes used during training contain 517 positives, 107 background non-vessels, and 118 ignores under this contract. Training never stages test or human-verified labels.

The frozen scorer greedily matches predictions and ground truth within 200 m, then aggregates precision, recall, and F1 across scenes. Each cell selects its threshold on development predictions and binds that operating point to the selected checkpoint. The public snapshot reports development-selection F1. It does not substitute that value for test F1, dark-vessel recall, or near-shore performance.

3 Related Work

xView3 frames dark-vessel detection as localization in Sentinel-1 SAR rather than box classification [3]. A CenterNet-style heatmap fits the point annotations and distance metric without inventing box extents [9]. We use this established detector form to hold the task head fixed.

The ViT track compares supervised ImageNet AugReg [4], optical SatDINO trained on fMoW-RGB [5], and SARMAE trained on SAR-1M [2]. SatDINO and SARMAE also differ in objective, so their gap is a released-checkpoint comparison rather than a pure modality intervention. The CNN track uses ConvNeXt-V2-Base [8]. Its optical and SAR arms come from matching BigEarthNet Sentinel-2 and Sentinel-1 land-cover models [6, 1]; this pair gives the cleaner modality contrast. ImageNet controls match the generic source dataset but not full training history across architectures.

4 Methods

Every arm predicts a stride-4 vessel heatmap from the same input. Gaussian targets mark positive points, and ignore neighborhoods mask LOW-confidence labels. The detector decodes local peaks above a permissive candidate floor, maps them to scene coordinates, and applies distance-based non-maximum suppression before scoring. ViT and CNN adapters reconcile native feature strides with the shared 256-channel head.

One PyTorch Lightning module and data module serve all cells. AdamW uses a 10^{-4} learning rate, 0.05 weight decay, layer-wise decay 0.65, five warm-up epochs, and cosine decay across at most 50 epochs. Full-scene development evaluation occurs every five epochs; early stopping waits four evaluation cycles. All runs use seed 0, micro-batch 16, accumulation 1, and end-to-end fine-tuning.

The corrected campaign runs one process per NVIDIA H100 in Lightning `32-true`. CUDA matrix-multiply TF32 and cuDNN TF32 stay disabled. Runtime receipts bind code, configuration, environment, GPU identity, checkpoint hashes, and completion markers. The result importer rejects a non-H100 cell, a duplicate identifier, a non-finite metric, a missing marker, or any fraction that lacks all eight completed arms.

5 Results and Analysis

The committed snapshot is an operational status export rather than a verified reverse-result bundle. It records the state of all 32 planned cells, but it does not carry the completion-marker and checkpoint hashes required to admit a metric. The fail-closed generator therefore emits no numerical comparison. Table 2 and Fig. 2 summarize campaign progress. The full inventory appears in Appendix A; dashes mean unavailable evidence, not zero performance.

Table 2: H100 campaign evidence at the frozen manuscript cutoff.

Done	Started	Pending	Contract verified	Reportable fractions
13	8	11	0	None

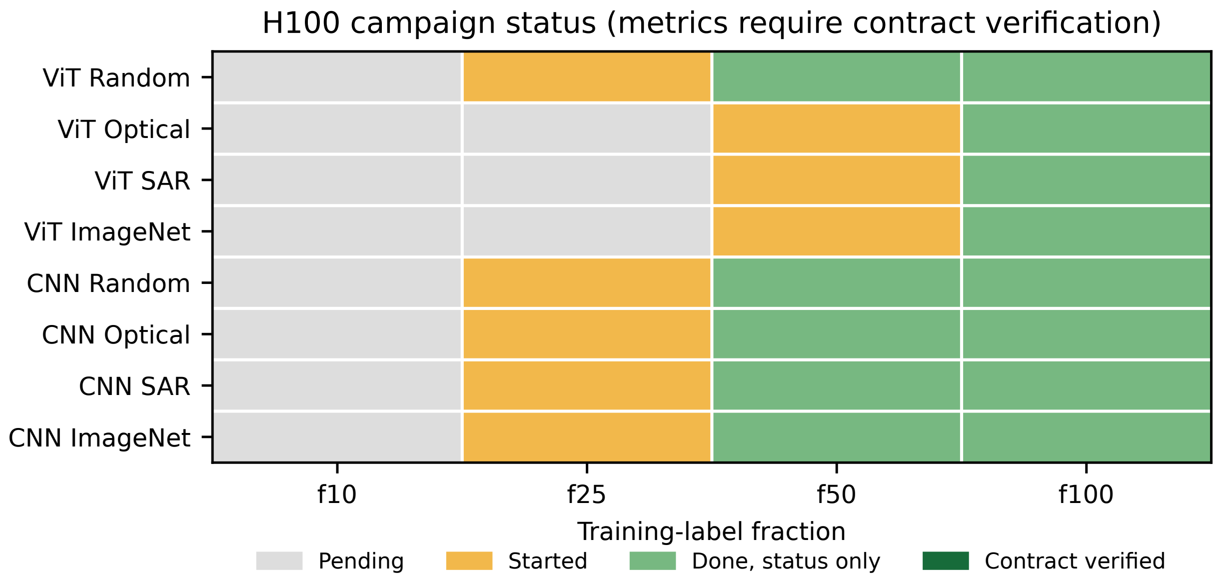

Figure 2: H100 campaign status at the manuscript cutoff. Status-only completions are not treated as reportable metrics.

This cutoff supports conclusions about execution coverage only. The full-data jobs have reached a terminal status in the operator snapshot, while work at the lower budgets continues. We do not treat terminal status alone as proof of a reportable checkpoint. A valid metric also needs its completion receipt, checkpoint binding, strict-FP32 provenance, and finite corrected-development record.

The status profile cannot establish an initialization ranking or label efficiency. A label-fraction comparison requires all eight roles at a common budget, and the generator must verify every corresponding receipt. We therefore make no claim about labels saved, curve monotonicity, scarce-label ordering, or transfer gains. One seed also precludes error bars and significance tests.

The complete analysis will make those same comparisons after the reverse bundle passes validation. The released checkpoints differ in objective and source-data volume, so these contrasts do not identify a causal effect of source modality alone.

6 Conclusion and Future Work

We built a controlled two-track experiment for measuring transfer under four nested label budgets. One detector, one corrected label contract, and one strict-FP32 H100 recipe constrain all 32 cells. The current deadline snapshot reports campaign progress but lacks the result receipts needed for a numerical claim. We therefore publish no H100 ranking or label-efficiency conclusion at this cutoff.

The next analysis step imports the verified reverse bundle, closes all 32 cells, and regenerates both papers from that single source. The team will then freeze the cohort, score the 16-scene test split, and use the 50 human-verified scenes once for dark-vessel and near-shore analysis. John Roth and Kyle Wagner jointly designed the study, implemented its data, model, training, and evaluation code, ran the experiments, analyzed the evidence, and wrote this report.

References

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A Full H100 Campaign Inventory

Table 3: H100 core-grid status. Development F1 is shown only for contract-verified completions; dashes are intentionally non-reportable.

Track	Initialization	Fraction (scenes)	Status	Evidence	Epoch	Duration	Dev F1
CNN	Random	10% (12)	Pending	status only	—	—	—
CNN	Random	25% (28)	Started	status only	7	2h07m	—
CNN	Random	50% (56)	Done	status only	49	1d02h	—
CNN	Random	100% (111)	Done	status only	49	2d03h	—
CNN	ImageNet	10% (12)	Pending	status only	—	—	—
CNN	ImageNet	25% (28)	Started	status only	0	10m	—
CNN	ImageNet	50% (56)	Done	status only	39	21h22m	—
CNN	ImageNet	100% (111)	Done	status only	49	2d03h	—
CNN	Optical	10% (12)	Pending	status only	—	—	—
CNN	Optical	25% (28)	Started	status only	1	18m	—
CNN	Optical	50% (56)	Done	status only	49	1d02h	—
CNN	Optical	100% (111)	Done	status only	49	2d03h	—
CNN	SAR	10% (12)	Pending	status only	—	—	—
CNN	SAR	25% (28)	Started	status only	0	15m	—
CNN	SAR	50% (56)	Done	status only	44	1d00h	—
CNN	SAR	100% (111)	Done	status only	24	1d01h	—
ViT	Random	10% (12)	Pending	status only	—	—	—
ViT	Random	25% (28)	Started	status only	0	0m	—
ViT	Random	50% (56)	Done	status only	49	15h01m	—
ViT	Random	100% (111)	Done	status only	34	19h31m	—
ViT	ImageNet	10% (12)	Pending	status only	—	—	—
ViT	ImageNet	25% (28)	Pending	status only	—	—	—
ViT	ImageNet	50% (56)	Started	status only	9	2h37m	—
ViT	ImageNet	100% (111)	Done	status only	49	1d03h	—
ViT	Optical	10% (12)	Pending	status only	—	—	—
ViT	Optical	25% (28)	Pending	status only	—	—	—
ViT	Optical	50% (56)	Started	status only	30	8h53m	—
ViT	Optical	100% (111)	Done	status only	49	1d03h	—
ViT	SAR	10% (12)	Pending	status only	—	—	—
ViT	SAR	25% (28)	Pending	status only	—	—	—
ViT	SAR	50% (56)	Started	status only	18	5h25m	—
ViT	SAR	100% (111)	Done	status only	44	1d00h	—

B Reproducibility and Audit Record

The public snapshot records its capture time, H100 hardware class, source SHA, Python and PyTorch versions, strict-FP32 backend state, frozen artifact hashes, and one row for every planned cell. The generator validates those fields before writing a manuscript artifact. The committed public log contains sanitized campaign progress and no acceptance-test evidence. A future submission may include acceptance artifacts after the importer verifies a result handback. The submission excludes imagery, labels, weights, checkpoints, run trees, credentials, and private paths.

The detector uses 512-pixel crops, a stride-4 heatmap, Gaussian standard deviation two output pixels, a candidate floor of 0.05, and 120 m distance NMS. Training uses at most 50 epochs with development evaluation every five epochs. The project fixes seed 0 and reports point estimates without uncertainty bands.

C Qualitative-Evidence Boundary

The snapshot used for this report contains status metadata rather than exported H100 scene overlays or contract-verified development metrics. We omit qualitative examples instead of substituting engineering probes or opening sealed imagery. After we freeze the cohort and analysis code, we will sample successes and errors from the pre-registered held-out outputs.